SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0510**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Scenario-Based Questions

Code of Conduct:

*I **AKHILA POCHIMIREDDY(192425225)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Akhila.p

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks (100)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

(a) Clearly interpret the scenario and explain the cause of low contrast.

A low-contrast image has only a small difference between the intensity values of the object and the background. As a result, the image appears dull and important features are difficult to distinguish. This usually occurs due to poor lighting, improper camera settings, limited dynamic range, or incorrect exposure.

(b) Identify all key issues affecting image visibility.

- Poor visibility of objects.
- Difficulty in distinguishing foreground and background.
- Loss of important image details.
- Reduced accuracy of segmentation and feature extraction.
- Poor visual quality.

(c) Apply an appropriate enhancement technique to address the problem.

Histogram Equalization redistributes pixel intensity values across the full intensity range, thereby improving the overall contrast of the image.

(d) Justify your choice with logical reasoning.

Histogram Equalization is effective because it enhances global contrast without altering the image content. It improves visibility, makes hidden features more distinguishable, and increases the performance of subsequent image processing tasks such as segmentation and edge detection.

(e) Present your explanation in a clear and structured manner.

The image suffers from low contrast due to limited intensity variation. Applying Histogram Equalization spreads intensity values over the entire range, enhancing visibility and improving image quality for further processing.

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

(a) Interpret the scenario and explain the impact of uneven illumination.

Uneven illumination occurs when different regions of an image receive different lighting intensities, producing bright and dark patches. This causes inconsistent brightness across the image.

(b) Identify key factors causing intensity variation.

- Non-uniform lighting source.
- Shadows.
- Reflections.
- Sensor illumination variations.
- Environmental lighting conditions.

(c) Apply a suitable enhancement method to normalize illumination.

CLAHE enhances contrast locally by dividing the image into small regions and applying histogram equalization separately to each region.

(d) Justify your selected approach with reasoning.

CLAHE improves local contrast while preventing over-enhancement. It effectively corrects uneven lighting, preserves image details, and avoids excessive brightness amplification.

(e) Provide a clear and well-structured explanation.

Uneven illumination reduces image consistency and visibility. CLAHE normalizes local brightness, improves contrast, and preserves important image details, making it suitable for images with varying illumination.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

(a) Explain the scenario and characteristics of this noise.

Salt-and-pepper noise appears as randomly distributed white (salt) and black (pepper) pixels. It commonly occurs during image acquisition, transmission, or scanning due to sensor or communication errors.

(b) Identify key issues impacting image quality.

- Reduced readability.
- Loss of fine image details.
- Difficulty in segmentation.
- Decreased image quality.

- Increased false edge detection.

(c) Apply an appropriate filtering technique.

The Median Filter replaces each pixel with the median value of its neighboring pixels, effectively removing impulse noise.

(d) Justify why this method is more suitable than alternatives.

Unlike mean filtering, the Median Filter removes salt-and-pepper noise while preserving edges and fine details. It is specifically designed for impulse noise removal and does not blur the image significantly.

(e) Present your response clearly and logically.

Salt-and-pepper noise introduces random black and white pixels that degrade image quality. Applying a Median Filter effectively removes this noise while maintaining important edges and image details.

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

(a) Interpret the scenario and explain why edges are blurred.

Blurred edges occur when excessive smoothing is applied during noise removal. High-frequency components corresponding to edges are weakened, resulting in loss of sharpness.

(b) Identify key issues related to smoothing and edge loss.

- Loss of edge information.
- Reduced image sharpness.
- Poor object boundary detection.
- Decreased segmentation accuracy.
- Reduced feature extraction performance.

(c) Apply an appropriate technique to restore edges.

The Laplacian Filter enhances edge information by emphasizing rapid intensity changes.

(d) Justify your decision with clear reasoning.

The Laplacian Filter restores sharpness by highlighting image edges without significantly affecting uniform regions. It improves boundary visibility and enhances feature extraction.

(e) Structure your explanation clearly.

Excessive smoothing removes important edge information. Applying the Laplacian Filter restores edge sharpness, improves object boundaries, and enhances overall image clarity

5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

(a) Interpret the scenario and describe high-frequency noise.

High-frequency noise consists of rapid intensity variations that appear as fine-grained noise in satellite images. It obscures important image features and reduces visual clarity.

(b) Identify the key issues affecting image clarity.

- Poor feature visibility.
- Reduced image clarity.
- Difficulty in identifying objects.
- Increased false edge detection.
- Lower segmentation accuracy.

(c) Apply an appropriate frequency domain filtering method.

The Gaussian Low-Pass Filter suppresses high-frequency noise while preserving significant image structures.

(d) Justify the choice of filter with reasoning.

A Gaussian Low-Pass Filter effectively reduces high-frequency noise without introducing ringing artifacts. Compared to the Ideal Low-Pass Filter, it produces smoother transitions and preserves important image information.

(e) Present your answer in a clear and organized manner.

High-frequency noise affects image interpretation by masking useful information. Applying a Gaussian Low-Pass Filter reduces noise while preserving essential image features, making it ideal for satellite image enhancement.